Day 1 · HTML Intro & Structure

· What is HTML?

→ Stands for HyperText Markup Language.

→ Gives skeleton structure to a webpage
(વેબસાઇટ નું કંકાલ / હાડપિંજર - Skeleton of a website)

> 🏠 *Real-life: HTML is like the bricks and walls of a house, CSS is paint and interior.*

```
<p>My first web page</p>
```
:- My first web page (shown on screen)

Remember: HTML is not a programming language; it has no logic or loops, only tags!

Mistake: Thinking HTML alone can make a beautiful website. It only makes the raw skeleton!

· Basic HTML Document Structure

→ Every HTML file must follow this exact hierarchy:

```
<!DOCTYPE html> → Tells browser it's HTML5
<html> → Root element of page
  ├── <head> → Metadata, title, fonts (invisible to user)
  └── <body> → Visible content (headings, paragraphs, images)
</html>
```

→ <head> = મગજ (brain - settings), <body> = શરીર (visible body) (મગજ અને શરીર)

```
<title>My Site</title>
```
:- Shows on browser tab at top

Remember: Only things inside <body> are shown on the screen!

Mistake: Putting <h1> or <p> inside <head>. Head is ONLY for metadata and title!

Day 2 · Headings & Paragraphs

· HTML Headings (<h1> to <h6>)

→ Six levels of headings from most important (<h1>) to least (<h6>).

→ Search engines like Google use <h1> to understand the main topic (મુખ્ય શીર્ષક - Main Heading)

> *Real-life: <h1> is newspaper headline, <h2> is sub-section, <h6> is small footnote.*

```
<h1>Main News</h1> ... <h6>Fine Print</h6>
```

:-

Main News

...

Fine Print

Remember: Use only ONE <h1> per page for best SEO! Never skip heading levels.

Mistake: Using <h1> just to make text bold or big. Use CSS for font size, not heading tags!

· Paragraphs (<p>),
 & <hr>

→ **<p>** defines a paragraph of text. Browser adds empty space before and after it.

→ **
** breaks line without extra gap (નવી લીટી - single line break)

→ **<hr>** draws a horizontal dividing line (આડી લીટી - thematic divider)

```
<p>Line 1<br>Line 2</p><hr>
```

:- Line 1
Line 2

Remember:
 and <hr> are self-closing void tags (no </br> needed).

Mistake: Using multiple

 tags to push items down instead of CSS margin!

Day 3 · Formatting Elements

· Text Formatting Tags

→ **<b>** vs ****: Both look bold, but adds semantic importance for screen readers (મહત્વપૂર્ણ અક્ષરો)

→ **<i>** vs ****: Both look italic, but adds verbal stress/emphasis (ત્રાંસા અક્ષરો)

```
<b>Bold</b> & <i>Italic</i>
```
:- **Bold** & *Italic*

Remember: Prefer and over <b> and <i> for accessibility & SEO.

Mistake: Overusing formatting tags on whole paragraphs instead of specific keywords.

· Special Markups: <mark>, <del>, <ins>, <sub>, <sup>

→ **<mark>**: Yellow highlighter effect (હાઈલાઈટ કરવું)

→ **<del>** and **<ins>**: Strikethrough & underline for price discounts/edits (જૂનો ભાવ અને નવો ભાવ)

→ **<sub>** (H_2O) & **<sup>** (x^2): Subscript & Superscript (નીચે અને ઉપર લખાણ)

```
<del>500</del> <ins>399</ins> | H<sub>2</sub>O
```
:- ~~500~~ 399 | H_2O

Remember: Use <del> and <ins> together for showing discounted prices in e-commerce.

Mistake: Confusing <sub> (submarine = goes down) with <sup> (superman = flies up).

Day 4 · Style Attribute & CSS Methods

· The style Attribute (Inline CSS)

→ Used to style a single specific HTML element directly inside opening tag.

→ Syntax: `style="property: value;"` (ટેગ ની અંદર જ સ્ટાઈલ આપવી)

> 👕 *Real-life: Wearing custom badge on one shirt vs uniform for whole school.*

```
<p style="color: red; font-size: 20px;">Alert</p>
```
:- Alert (in red color)

Remember: Always end each CSS property inside style with a semicolon (;)!

Mistake: Writing huge styles in HTML tags. It makes code messy and hard to maintain.

· 3 Ways to Connect CSS to HTML

→ 1. **Inline CSS:** inside tag via `style="..."`

→ 2. **Internal CSS:** inside `<style>` tag within <head> (પેજ ના માથામાં)

→ 3. **External CSS:** separate `.css` file connected via `<link>` in <head>
(અલગ ફાઈલ જોડીને)

```
<link rel="stylesheet" href="style.css">
```
:- Connects external styling file

Remember: External CSS is best practice for professional production websites!

Mistake: Placing <link> tag inside <body>. It MUST always be inside <head>!

Day 5 · HTML Linking (<a> tag)

· Hyperlinks & href Attribute

→ **<a>** (Anchor tag) creates clickable links to other pages or websites.

→ **href:** Hypertext Reference (the target URL) (લિંક નું સરનામું - Destination URL)

→ **target="_blank":** Opens link in a brand new browser tab (નવા ટેબમાં ખોલવું)

```
<a href="https://google.com" target="_blank">Google</a>  :- Clickable link opening Google
```

Remember: When using target="_blank", always add rel="noopener noreferrer" for security!

Mistake: Leaving href empty or writing <a> without href attribute (it won't be clickable).

· Special Links: Email, Phone & Bookmarks

→ **mailto:** Opens default email client (ઈમેલ મોકલવા)

→ **tel:** Prompts phone dialer on mobile devices (ફોન કોલ કરવા)

→ **#id:** Jumps to specific section on the same page (પેજ ની અંદર કૂદકો મારવો)

```
<a href="mailto:a@b.com">Mail</a> | <a href="#sec1">Jump</a>  :- Opens email app | Jumps to #sec1
```

Remember: Bookmark links match the `id` of the target tag (e.g. href="#top").

Mistake: Forgetting the '#' symbol in internal bookmark link hrefs!

"Connection is the energy that exists between people when they feel seen and heard"

Day 6 · Images (<img> tag)

· Image Embedding & Attributes

→ **<img>** is an empty self-closing tag used to display images on webpage.

→ **src:** Source path (URL or local folder path) of image (ફોટા નું સરનામું)

→ **alt:** Alternative text displayed if image fails to load (ફોટો ન દેખાય તો લખાણ)

```
<img src="logo.png" alt="College Logo" width="120">
```
:- Displays 120px wide image

Remember: Never omit the `alt` attribute! It is critical for blind users & Google SEO.

Mistake: Writing wrong folder path (like missing images/ or typo in image extension .png vs .jpg).

· Responsive Images & loading="lazy"

→ **width & height:** Reserve layout space to prevent page jumping during load.

→ **loading="lazy":** Defers loading image until user scrolls near it
(પેજ ઝડપી બનાવવું - speed up page)

→ Always scale proportionally using CSS `max-width: 100%; height: auto;`

```
<img src="banner.jpg" loading="lazy" alt="Banner">
```
:- Loads only when scrolled into view

Remember: Use loading="lazy" on all images below the fold to save user bandwidth!

Mistake: Distorting image aspect ratio by setting both fixed width and fixed height in px.

Day 7 · HTML Tables

· Table Basics: <table>, <tr>, <th>, <td>

→ **<table>:** Container for tabular data (rows & columns).

→ **<tr>:** Table Row (ટેબલ ની આડી હરોળ)

→ **<th>:** Table Header (bold & centered by default) (કોલમ નું હેડિંગ)

→ **<td>:** Table Data cell (regular content) (ટેબલ નો ડેટા ખાનું)

```
<table border='1'>
  <tr><th>Name</th><th>Roll</th></tr>
  <tr><td>Ved</td><td>101</td></tr>
</table>
```

Remember: A table is constructed strictly ROW BY ROW! Every row starts with <tr>.

Mistake: Putting text directly inside <table> without wrapping in <tr> and <td>.

· Merging Cells: colspan & rowspan

→ **colspan="N":** Merges N columns horizontally into one cell (આડા ખાના ભેગા કરવા)

→ **rowspan="N":** Merges N rows vertically into one cell (ઊભા ખાના ભેગા કરવા)

→ Semantic tags: <thead>, <tbody>, <tfoot> group header, body, footer.

```
<td colspan="2">Full Address</td>
```
:- One cell spanning 2 columns

Remember: When you merge cells using colspan=2, delete the adjacent <td> in that row!

Mistake: Using tables for overall website layouts! Tables are ONLY for tabular data.

"Order and structure are the foundations of all good work"

Day 8 · Lists in HTML

· Unordered (<ul>) & Ordered (<ol>) Lists

→ **<ul>:** Bulleted list (dots, circles, squares) (બુલેટ લિસ્ટ - ક્રમ વગર)

→ **<ol>:** Numbered list (1, A, a, I, i via type attribute) (નંબર વાળું લિસ્ટ)

→ **<li>:** List Item (must be inside <ul> or <ol>) (દરેક આઈટમ)

```
<ol type="A" start="3"><li>Item</li></ol>
```
:- C. Item (starts numbering from C)

Remember: Only <li> elements can be direct children of <ul> and !

Mistake: Putting <p> or directly inside <ul> instead of inside <li>.

· Description Lists (<dl>) & Nested Lists

→ **<dl>:** Description list for terms & definitions (like a glossary).

→ **<dt>:** Definition Term (શબ્દ / ટર્મ)

→ **<dd>:** Definition Description (automatically indented) (શબ્દ નો અર્થ / સમજૂતી)

→ **Nested lists:** Placing a new <ul> inside an existing <li> creates sub-menus.

```
<dl><dt>HTML</dt><dd>Markup Lang</dd></dl>
```
:- HTML
Markup Lang

Remember: Nested lists are the foundation of all website navigation dropdown menus!

Mistake: Placing the sub-list between and <li> instead of INSIDE the parent <li>.

"Organization isn't about perfection. It is about efficiency and peace of mind"

Day 9 · Block vs Inline Elements

· Block-level Elements (, <p>, <h1>)

→ Always starts on a new line and takes up the full width available (100%).

→ Can have margin, padding, width, and height set in CSS. (આખી લાઈન રોકી લે)

→ Examples: , <p>, <h1>-<h6>, <ul>, <li>, <section>, <form>

[------- Block Element takes 100% width of parent row -------]

Remember: is the universal generic block container for grouping elements.

Mistake: Wrapping block elements inside inline elements (like <h1> inside).

· Inline Elements (, <a>, <b>)

→ Does NOT start on a new line; takes only as much width as its content.
(જેટલી જરૂર હોય તેટલી જ જગ્યા)

→ Top & bottom margin/padding do NOT push adjacent lines away!

→ Examples: , <a>, <b>, <i>, <img>, <code>,

→ **:** Generic inline container for styling specific words in a sentence.

Price: $99 :- Price: $99 (stays in same line)

Remember: Use for building boxes/sections; use for coloring words inside text.

Mistake: Trying to apply CSS width and height to inline elements (they will be ignored!).

Day 10 · Iframes & Computer Code Tags

· HTML <iframe> (Inline Frame)

→ Used to embed another document or website inside the current page.
(બીજી સાઈટ અંદર બતાવવી)

→ Commonly used for Google Maps, YouTube video players, and widgets.

→ Key attributes: `src`, `width`, `height`, `loading="lazy"`, `sandbox`

> *Real-life: An iframe is like a glass window inside your room looking into another room.*

```
<iframe src="map.html" width="300" height="200"></iframe>
```
:- Embeds 300x200 window

> **Remember:** Always use the `sandbox` attribute to prevent malicious scripts from external iframes!

> **Mistake:** Many modern sites (like Google, Facebook) block being embedded in iframes using X-Frame-Options.

· Computer Code Tags: <code>, <pre>, <kbd>, <samp>

→ **<code>:** Displays code in monospace font (કોડ લખાણ)

→ **<pre>:** Preserves exact spaces and line breaks as typed (જગ્યા અને લાઈન સાચવી રાખે)

→ **<kbd>:** Represents keyboard keys (કીબોર્ડ ની કી - e.g. Ctrl + C)

→ **<samp>:** Represents sample computer terminal output

```
Press <kbd>Ctrl</kbd> + <kbd>C</kbd> to copy
```
:- Press `Ctrl` + `C` to copy

> **Remember:** Always wrap multi-line code inside <pre><code> ... </code></pre>.

> **Mistake:** Expecting <code> alone to preserve spaces; without <pre>, spaces get collapsed into one!

"Programs must be written for people to read, and only incidentally for machines to execute"

Day 11 · Semantic HTML Elements

· What is Semantic HTML?

→ Semantic elements clearly describe their meaning to browser and developer.
(અર્થપૂર્ણ ટેગ્સ - tags with meaning)

→ Non-semantic: `<div>`, `<span>` (tells nothing about content).

→ Semantic: `<header>`, `<nav>`, `<main>`, `<footer>`

```
+--------------------------------------+
| <header> Logo & Site Title |
| <nav> Home | About | Contact |
+------------------+-------------------+
| <main> | <aside> Sidebar |
| <section> | Trending links |
| <article> | |
+------------------+-------------------+
| <footer> Copyright © 2026 |
+--------------------------------------+
```

Remember: Semantic tags boost SEO ranking and allow screen readers to navigate effortlessly.

Mistake: Building an entire website using 100 tags ('div soup') instead of semantic tags!

· <section>, <article> & <figure>

→ **<section>:** Thematic grouping of content, usually with a heading.

→ **<article>:** Self-contained, independently distributable post/card
(સ્વતંત્ર લેખ - blog post/news)

→ **<aside>:** Sidebar content tangentially related to main content.

→ **<figure> & <figcaption>:** Self-contained photo with caption underneath.

```
<figure><img src="cat.jpg"><figcaption>My Cat</figcaption>
</figure>
```
:- Image paired with caption

Remember: An <article> should make sense on its own if sent via RSS feed or newsletter.

Mistake: Using <section> as a plain styling wrapper; use for purely visual styling.

Day 12 · Entities & Symbols

· HTML Reserved Character Entities

→ Some characters are reserved in HTML (like < and >) because browsers treat them as tags. (ખાસ ચિન્હો - reserved characters)

→ To display these characters on screen, use entity names starting with & and ending with ;

→ **<** = < (less than) | **>** = > (greater than)

→ **&** = & (ampersand) | **"** = " (double quote)

To write <p>, code: <p> :- Shows <p> literally on screen

Remember: Always use   when you need a non-breaking space between two words!

Mistake: Typing literal '<' in paragraph text; older browsers may interpret it as an unclosed tag!

· Symbols & Emoji Codes

→ **Copyright:** © :- © | **Registered:** ® :- ®

→ **Trademark:** ™ :- ™ | **Euro / Pound:** € £ :- € £

→ **Emojis:** Encoded via UTF-8 decimal code (e.g. 😀 = 😀) (ઈમોજી કોડ)

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Remember: Ensure <meta charset="UTF-8"> is present in <head> so all symbols and emojis render!

Mistake: Forgetting the trailing semicolon (;) on entities like typing &copy without ';'!

"Every symbol carries a story, every character has a place"

Day 13 · Forms Part 1: Basics & Inputs

· The <form> Element & HTTP Methods

→ Used to collect user data and send it to server for processing.
(યુઝર ડેટા ફોર્મ - collecting user data)

→ **action:** URL where form data is submitted to.

→ **method:** HTTP method: GET (shows in URL) vs POST (secure, hidden body)

> *Real-life: GET is like a public postcard; POST is like a sealed, private envelope.*

`<form action="/login" method="POST"> ... </form>` :- Submits private payload securely

Remember: Use POST for passwords, credit cards, or large data; use GET for search forms!

Mistake: Using method="GET" for login passwords (it exposes password directly in URL history!).

· Core Input Types (<input>)

→ **type="text":** Single line text field (ટેક્સ્ટ બોક્સ)

→ **type="password":** Masks characters as dots/asterisks (પાસવર્ડ બોક્સ - ગુપ્ત અક્ષરો)

→ **type="email":** Automatically validates email format (@ symbol)

→ **type="number":** Numbers only with min, max, and step controls

`<input type="email" name="user_email" required>` :- Input field with email validation

Remember: Every input MUST have a name attribute, or its value will NOT be sent to server!

Mistake: Using type="text" for phone/passwords instead of specialized type="tel" or "password".

"Listening to user input is the starting point of every interactive system"

Day 14 · Forms Part 2: Controls & Options

· Radio, Checkbox & Dropdown

→ **type="radio":** Select ONLY ONE option from a group (must share same `name`).
(એક જ પસંદગી - Gender/Option)

→ **type="checkbox":** Select MULTIPLE options independently (એકથી વધુ પસંદગી)

→ **<select> & <option>:** Dropdown selection menu (ડ્રોપડાઉન લિસ્ટ)

```
<input type="radio" name="g" value="m"> Male
```
:- ( ) Male (radio selection dot)

Remember: Radio buttons must share the exact same `name` to make them mutually exclusive!

Mistake: Giving different name attributes to related radio buttons (user will be able to select all!).

· <label>, <textarea> & Attributes

→ **<label for="id">:** Clicking text focuses the corresponding input box.
(ટેક્સ્ટ પર ક્લિક કરવાથી બોક્સ સિલેક્ટ થાય)

→ **<textarea rows="4">:** Multi-line text input for comments/messages.

→ **Attributes:** `placeholder` (hint), `required` (mandatory), `disabled`, `value`

```
<label for="n">Name</label><input id="n" required>
```
:- Accessible label bound to input

Remember: Always pair every input with a <label> using `for` and `id` for accessibility!

Mistake: Using placeholder as a replacement for label. Placeholders disappear when user types.

"User experience is about making interactions effortless and self-evident"

Day 15 · Audio & Video Media

· Embedding Audio (<audio>)

→ Used to play sound files (mp3, wav, ogg) on webpage without external plugins.
(ઓડિયો / ગીત વગાડવું)

→ **controls:** Displays play, pause, and volume bar.

→ **<source src="..." type="audio/mpeg">:** Provides fallback audio formats.

```
<audio controls><source src="a.mp3"></audio>
```
:- Audio player with play button

Remember: Always include the `controls` attribute, otherwise the audio player is completely invisible!

Mistake: Using `autoplay` with sound; modern browsers block audio autoplay to protect users.

· Embedding Video (<video>)

→ Used to embed video clips (mp4, webm). (વિડીયો પ્લેયર - video on page)

→ **poster="img.jpg":** Thumbnail image displayed before video plays.
(વિડીયો શરૂ થતા પહેલા નો ફોટો)

→ **attributes:** `controls`, `autoplay muted`, `loop`, `width="640"`

```
<video controls poster="thumb.jpg" width="400"></video>
```
:- Video box with thumbnail

Remember: If you want video to autoplay on page load, you MUST add `muted`!

Mistake: Uploading huge raw 4K video files directly; compress to WebM or stream via YouTube/Vimeo.

Day 16 · Meta Tags & SEO Head

· Essential Meta Tags in <head>

→ Meta tags live inside <head> and supply metadata to browsers & search engines. (પેજ ની છુપી માહિતી - metadata)

→ **charset="UTF-8":** Supports all languages (English, Gujarati, Hindi, Emoji).

→ **viewport:** `content="width=device-width, initial-scale=1.0"` enables mobile responsiveness! (મોબાઈલ સ્ક્રીન સાઈઝ સેટિંગ)

```
<meta charset="UTF-8">
<meta name="viewport" content="width=device-width, initial-scale=1.0">
<meta name="description" content="Best HTML Study Notes">
```

Remember: Without the viewport meta tag, mobile phones will display desktop sites tiny and zoomed-out!

Mistake: Forgetting charset="UTF-8" leads to Gujarati or Hindi characters turning into garbage symbols ().

· SEO & Social Sharing Meta Tags

→ **meta description:** The 1-2 sentence snippet displayed in Google search results. (ગૂગલ સર્ચ રિઝલ્ટ લખાણ)

→ **Open Graph (og:title, og:image):** Controls title & preview card when shared on WhatsApp/Twitter. (વોટ્સએપ શેરિંગ કાર્ડ)

→ **<link rel="icon" href="favicon.ico">:** Displays small logo on browser tab.

```
<meta name="description" content="Free CSE Web Dev Notes">   :- Sets Google search snippet
```

Remember: Keep meta description between 150-160 characters so Google does not cut it off!

Mistake: Stuffing 50 repetitive keywords in meta keywords tag; Google ignores keyword tags today.